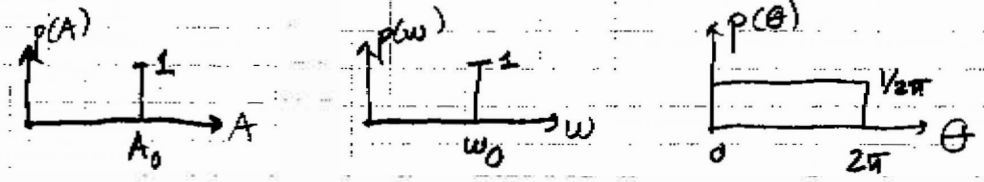
Problem 2.3

1) 

2)  ← 3 af medlemmerne i ensemblet

3) $\mu_X = E[X(t)] = E_\theta[A_0\cos(\omega_0 t+\theta)] = A_0\frac{1}{2\pi}\int_0^{2\pi}\cos(\omega_0 t+\theta)d\theta = 0$

4) $\hat{\mu}_x = A[X(t)] = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}X(t)dt = \lim_{T\to\infty}\frac{1}{2T}A_0\int_{-T}^{T}\cos(\omega_0 t+\theta)dt = 0$

5) $R_{XX}(\tau) = E[X(t)X(t+\tau)] = \frac{A_0^2}{2}E_\theta[\cos(\omega_0\tau)+\cos(2\omega_0 t+\omega_0\tau+2\theta)]$
$= \frac{A_0^2}{2}\cos\omega_0\tau + \frac{1}{2\pi}\int_0^{2\pi}\cos(2\omega_0 t+\omega_0\tau+2\theta)d\theta$
$= \frac{A_0^2}{2}\cos\omega_0\tau$

6) $\hat{R}_{XX}(\tau) = A[X(t)X(t+\tau)] = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}X(t)X(t+\tau)dt$
$= \lim_{T\to\infty}\frac{1}{2T}\frac{A_0^2}{2}\left[\int_{-T}^{T}\cos\omega_0\tau\,dt + \int_{-T}^{T}\cos(2\omega_0 t+\omega_0\tau+2\theta)dt\right]$
$= \frac{A_0^2}{2}\cos\omega_0\tau$

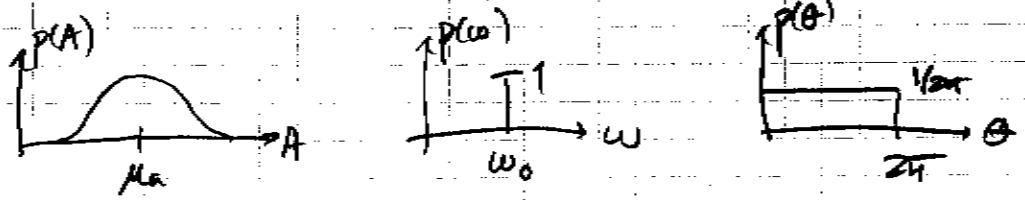
7) X(t) er WSS, da 1) $\mu_X(t) = \mu_X$ (konstant)
2) $R_{XX}(\tau)$ afhænger ikke af t

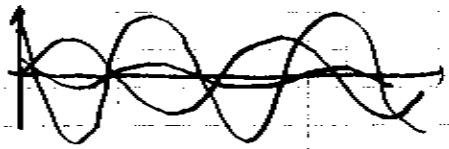
8) X(t) er ergodisk, da $\mu_X = \hat{\mu}_X$, $R_{XX}(\tau) = \hat{R}_{XX}(\tau)$

9) OK

10) $E[X^2(t)] = R_{XX}(0) = \frac{1}{2}A_0^2$ total effekt
$Var[X(t)] = E[X^2(t)] - \mu_X^2 = \frac{1}{2}A_0^2$ AC effekt
rms-værdi $= \sqrt{Var} = A_0/\sqrt{2}$

Problem 2.4

1) 

2) 

3) $\mu_X = E[X(t)] = E_{A,\Theta}[X(t)] = E_A[A]\,E_\Theta[\cos(\omega_0 t+\Theta)]$
$= \mu_a \frac{1}{2\pi}\int_0^{2\pi}\cos(\omega_0 t+\theta)\,d\theta = 0$

4) $\hat{\mu}_X = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}X(t)\,dt = 0$

5) $R_{XX}(\tau) = E_{A,\Theta}[X(t)X(t+\tau)]$
$= E_{A,\Theta}\left[\frac{A^2}{2}(\cos\omega_0\tau + \cos(2\omega_0 t+\omega_0\tau+2\Theta))\right]$
$= \frac{1}{2}E_A[A^2]\left\{E_\Theta[\downarrow] + E_\Theta[\downarrow]\right\}$
$= \frac{1}{2}(\mu_a^2+\sigma_a^2)\cos\omega_0\tau + 0$
$= \frac{1}{2}(\mu_a^2+\sigma_a^2)\cos\omega_0\tau$

6) $\hat{R}_{XX}(\tau) = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}X(t)X(t+\tau)\,dt$
$= \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\frac{1}{2}a^2(\cos(\omega_0\tau)+\cos(2\omega_0 t+\omega_0\tau+2\Theta))\,dt$
$= \frac{1}{2}a^2\cos\omega_0\tau$

7) WSS ja 1) μ_X konstant
2) $R_{XX}(\tau)$ afh. ikke af t

8) ergodisk nej $R_{XX}(\tau) \neq \hat{R}_{XX}(\tau)$

9) OK

10) $E[X^2(t)] = R_{XX}(0) = \frac{1}{2}(\mu_a^2+\sigma_a^2)$ total effekt
$Var(X(t)) = R_{XX}(0) - \mu_X^2 = R_{XX}(0)$ AC-effekt